

Slowing the Progression of Kidney Disease

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INTRODUCTION

Prior to 2002, there was no standardized definition for chronic kidney disease (CKD). Instead, terms such as *chronic renal failure or insufficiency*, *chronic renal disease*, and *progressive renal insufficiency* were used.¹ In 2002, the US National Kidney Foundation established the Kidney Disease Outcomes Quality Initiative (KDOQI)²; an early goal was to define CKD and classify its severity using nomenclature that could be scaled and spread. The KDOQI framework defined CKD as structural or functional abnormalities of the kidney present for at least 3 months, manifested either as kidney damage (most frequently detected as persistent albuminuria) with or without a glomerular filtration rate (GFR) less than 60 mL/min/1.73 m², or as decreased GFR with or without other evidence of kidney damage.² In 2012, Kidney Disease: Improving Global Outcomes (KDIGO) updated the framework to enhance its capacity for predicting risk of progression to kidney failure or development of cardiovascular disease (CVD).³ Central to this revision was to highlight the role of albuminuria as a marker of damage, an early manifestation of CKD, and an important prognostic factor for the progression of kidney disease and development of CVD.³

Developing a common framework for the presence and severity of CKD was instrumental in accelerating the pace of clinical research that led to better outcomes for people with kidney disease. In this chapter, we review techniques for evaluating and monitoring key indicators of CKD, describe factors associated with the risk of CKD progression, and outline measures to prevent adverse outcomes associated with CKD, such as kidney failure and CVD.

TECHNIQUES FOR EVALUATING AND MONITORING KEY INDICATORS OF CHRONIC KIDNEY DISEASE

Glomerular Filtration Rate

The key index of kidney function is the GFR, which cannot be measured directly but can be estimated from the urinary clearance of an ideal filtration marker. The most common methods used to estimate GFR are serum creatinine (SCr) concentration, creatinine clearance (CrCl), and estimation equations based on serum creatinine such as the Cockcroft-Gault (C-G) equation, and equations used in the Modification of Diet in Renal Disease (MDRD) study and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI).⁴ Details of GFR estimation are presented in [Chapters 3](#).

Albuminuria

The preferred method of measuring proteinuria or albuminuria is by calculating urine protein-to-creatinine ratio (PCR) or albumin-to-creatinine ratio (ACR) on a random urine specimen, respectively.³ Because it is highly accurate and convenient, ACR is preferred to the 24-hour urine collection for quantitative measurement of significant

urinary albumin. Random daytime specimens are acceptable if the preferred first morning specimens are not available. The relative merits of the PCR compared with ACR for detection and monitoring kidney damage are unclear.³ However, because albuminuria is a more sensitive marker than total protein in CKD due to diabetes, hypertension, and glomerular diseases, the ratio in adult spot urine samples should ideally be measured with ACR³; if ACR is high, PCR is acceptable. Methods are being developed to estimate ACR directly from PCR.⁵

EPIDEMIOLOGY AND BURDEN OF CHRONIC KIDNEY DISEASE

The number of individuals with kidney failure (end-stage kidney disease) represents less than 2% of those with CKD (stages 1–4). Although most people with CKD will not develop kidney failure, the minority that do pose a significant challenge to the health care system in terms of morbidity and cost, especially in low- and middle-income countries.

Globally, CKD is an important public health problem. According to estimates from the Global Burden of Disease Injuries and Risk Factors study, approximately 1.2 million people died from CKD worldwide in 2017, and the global all-age mortality rate linked to CKD increased over 40% between 1990 and 2017.⁶ In absolute terms, CKD is estimated to affect approximately 700 million people around the world, with a global prevalence of 9.1% (range, 8.5%–9.8%), and the global all-age prevalence of CKD has increased over 29% over the last 3 decades.⁶ CKD is associated with high risk of mortality and hospitalization, as well as high health care costs. Like diabetes, CKD has been described as a coronary artery disease “risk equivalent,” reflecting the high risk of vascular disease in this population.⁷

Susceptibility to CKD is higher in certain families and among members of certain races. This highlights the possibility of genetic predisposition to CKD. For instance, in Canada, the disproportionate burden of CKD among Indigenous populations may reflect the high prevalence of major drivers of CKD such as hypertension and diabetes; patterns are similar for Indigenous populations in Australia and New Zealand. Among the key risk factors for the initiation of CKD are hypertension, diabetes, hyperlipidemia, obesity, and smoking. In low- and middle-income countries, key risk factors for initiation of CKD may also include communicable diseases such as HIV, hepatitis B and C, malaria, schistosomiasis, and tuberculosis. This double burden of communicable and noncommunicable risk factors has led to a growing burden of CKD across all world regions and population subgroups. Further details on the epidemiology of CKD are presented in [Chapter 80](#).

RISK FACTORS FOR PROGRESSION OF CHRONIC KIDNEY DISEASE

The natural history of CKD progresses from susceptibility associated with the presence of risk factors ([Table 82.1](#)), to the development of

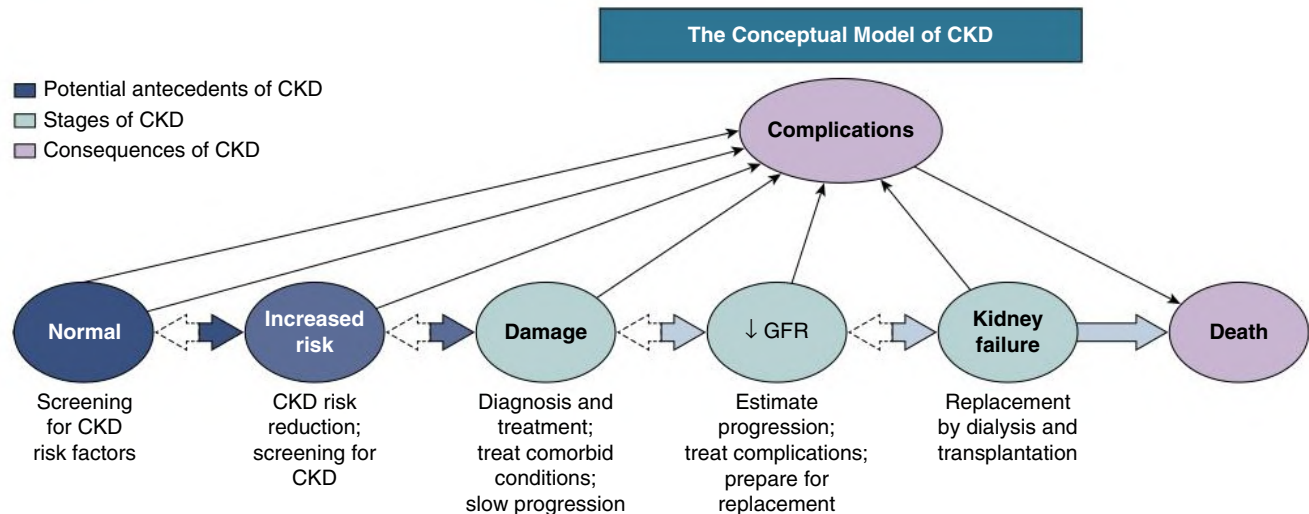


Fig. 82.1 Conceptual model of chronic kidney disease (CKD). *GFR*, Glomerular filtration rate. (Modified from Levey AS, Stevens LA, Coresh J. Conceptual model of CKD: applications and implications. *Am J Kidney Dis*. 2009;53[suppl 3]:S4–16.)

TABLE 82.1 Risk Factors Associated With the Initiation and Progression of Chronic Kidney Disease

Initiation Factors	Progression Factors
Systemic hypertension	Older age
Diabetes mellitus	Sex (male)
Cardiovascular disease	Race/ethnicity
Dyslipidemia	Genetic predisposition
Smoking	Poor BP control
Obesity/metabolic syndrome	Poor glycemic control
Hyperuricemia	Proteinuria
Low socioeconomic status	Cardiovascular disease
Nephrotoxin exposure: abuse of NSAIDs, analgesics, traditional herbal remedies; exposure to heavy metals, lead	Dyslipidemia
	Smoking
	Obesity/metabolic syndrome
	Hyperuricemia
	Low socioeconomic status
	Nephrotoxins (NSAIDs, contrast agents)
	Recurrent AKI events
	Metabolic acidosis

AKI, Acute kidney injury; CKD, chronic kidney disease; NSAIDs, nonsteroidal antiinflammatory drugs.

CKD and progression to kidney failure requiring renal replacement therapy, CVD, and/or death (Fig. 82.1).

The risk of progressive CKD is variable and affected by several factors, including demographic characteristics, clinical factors, and access to care (see Table 82.1).⁸ Epidemiologic data suggest that the natural history of CKD stages 3 to 4 and development of other adverse health outcomes vary based on patients' baseline risk factor profiles (Fig. 82.2).⁸ Many CKD patients with GFR less than 60 mL/min/1.73 m² do not reach stage 5 (kidney failure); among those with disease progression, the process is often not linear, with breakpoints in the progression pattern due to a number of factors, including the natural progression of underlying conditions such as infections, dehydration, CKD risk factors (blood pressure [BP], proteinuria), exposure to nephrotoxins (drugs, contrast agents), and acute kidney injury (AKI).

The rate of progression of CKD also varies based on the severity of underlying disease and between individual patients. Historically, the rate of decline in GFR of patients with diabetic nephropathy has been among the fastest, averaging around −10 mL/min/yr. Control of hypertension slows the rate of GFR decline to −5 mL/min/yr, with further improvement (−1 to −2 mL/min/yr) expected in patients with optimally controlled glycemia and hypertension. In nondiabetic nephropathy, patients with chronic glomerulonephritis exhibit a rate of CKD progression that is more than twice as fast as patients with chronic interstitial nephritis and 1.5 times faster than patients with hypertensive nephrosclerosis. The association of albuminuria and faster progression of CKD has been highlighted in a number of studies. Optimal hydration, relief of obstruction (where indicated), discontinuation of nephrotoxic agents, and control of hypertension often stabilize kidney function in a large percentage of patients. Patients with polycystic kidney disease and impaired kidney function (CKD stage 3b and beyond) may also have a faster rate of progression compared with other nephropathies. More rapid progression of CKD is linked to certain genetic, demographic, and clinical risk factors (see Table 82.1). Recently, the notion of a linear progression to the onset of kidney failure is being challenged with the proposition that the progression of CKD is strongly influenced by repeated AKI episodes, and that this is most often a hemodynamically driven process.⁹

SLOWING PROGRESSION OF CHRONIC KIDNEY DISEASE

Since the publication of the hyperperfusion-hyperfiltration hypothesis, considerable research has focused on understanding the pathophysiology of CKD,¹⁰ mechanisms of progression, and prevention of kidney failure (Fig. 82.3). Clinical strategies were initially based on manipulation of dietary protein intake and evolved to include inhibition of the renin-angiotensin-aldosterone system (RAAS) and control of glycemia and dyslipidemia (Table 82.2),¹¹ along with appropriate follow-up based on the level of estimated glomerular filtration rate (eGFR) and severity of albuminuria (Fig. 82.4).

A key goal of clinical follow-up is to recognize reversible causes of acute-on-chronic kidney disease (mimicking progression), such as acute illnesses with volume depletion. Patients with significant comorbidity burden—in particular, chronic heart disease, older age, and use of specific medications (e.g., RAAS inhibitors, sodium-glucose

Associations of CKD Markers With Adverse Health Outcomes

All-cause mortality					Cardiovascular mortality				
	ACR <10	ACR 10–29	ACR 30–299	ACR ≥300		ACR <10	ACR 10–29	ACR 30–299	ACR ≥300
eGFR >105	1.1	1.5	2.2	5.0	eGFR >105	0.9	1.3	2.3	2.1
eGFR 90–105	Ref	1.4	1.5	3.1	eGFR 90–105	Ref	1.5	1.7	3.7
eGFR 75–90	1.0	1.3	1.7	2.3	eGFR 75–90	1.0	1.3	1.6	3.7
eGFR 60–75	1.0	1.4	1.8	2.7	eGFR 60–75	1.1	1.4	2.0	4.1
eGFR 45–60	1.3	1.7	2.2	3.6	eGFR 45–60	1.5	2.2	2.8	4.3
eGFR 30–45	1.9	2.3	3.3	4.9	eGFR 30–45	2.2	2.7	3.4	5.2
eGFR 15–30	5.3	3.6	4.7	6.6	eGFR 15–30	14	7.9	4.8	8.1

Kidney failure (ESKD)					Acute kidney injury (AKI)					Progressive CKD				
	ACR <10	ACR 10–29	ACR 30–299	ACR ≥300		ACR <10	ACR 10–29	ACR 30–299	ACR ≥300		ACR <10	ACR 10–29	ACR 30–299	ACR ≥300
eGFR >105	Ref	Ref	7.8	18	eGFR >105	Ref	Ref	2.7	8.4	eGFR >105	Ref	Ref	0.4	3.0
eGFR 90–105	Ref	Ref	11	20	eGFR 90–105	Ref	Ref	2.4	5.8	eGFR 90–105	Ref	Ref	0.9	3.3
eGFR 75–90	Ref	Ref	3.8	48	eGFR 75–90	Ref	Ref	2.5	4.1	eGFR 75–90	Ref	Ref	1.9	5.0
eGFR 60–75	Ref	Ref	7.4	67	eGFR 60–75	Ref	Ref	3.3	6.4	eGFR 60–75	Ref	Ref	3.2	8.1
eGFR 45–60	5.2	22	40	147	eGFR 45–60	2.2	4.9	6.4	5.9	eGFR 45–60	3.1	4.0	9.4	57
eGFR 30–45	56	74	294	763	eGFR 30–45	7.3	10	12	20	eGFR 30–45	3.0	19	15	22
eGFR 15–30	433	1044	1056	2286	eGFR 15–30	17	17	21	29	eGFR 15–30	4.0	12	21	7.7

Fig. 82.2 Associations of chronic kidney disease (CKD) markers with adverse health outcomes. *ACR*, Albumin-to-creatinine ratio; *AKI*, acute kidney injury; *eGFR*, estimated glomerular filtration rate; *ESKD*, end-stage kidney disease. (Modified from Levey AS, de Jong PE, Coresh J, et al. The definition, classification, and prognosis of chronic kidney disease: a KDIGO Controversies Conference report. *Kidney Int.* 2011;80[1]:17–28.)

Timeline of Innovations in CKD Evaluation and Management

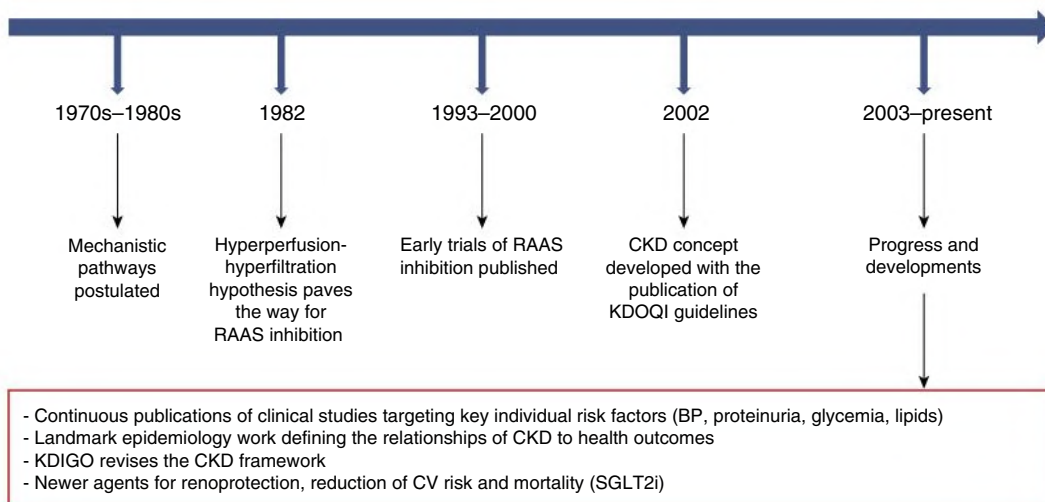


Fig. 82.3 Timeline of innovations in chronic kidney disease (CKD) evaluation and management. *BP*, Blood pressure; *CV*, cardiovascular; *KDIGO*, Kidney Disease: Improving Global Outcomes; *KDOQI*, Kidney Disease Outcomes Quality Initiative; *RAAS*, renin-angiotensin-aldosterone system; *SGLT2i*, sodium-glucose cotransporter-2 inhibitor.

TABLE 82.2 Evidence-Based Strategies to Slow the Progression of Kidney Disease

Risk Factor	Guideline Concordant Treatment Goals and Recommended Agents	Certainty of Evidence	Strength of Recommendation
Overweight	Maintain a healthy weight (BMI 20–25 kg/m ²)	D	1
Diet	Lower or maintain salt intake to <90 mmol/day (equivalent to <2 g sodium/day or <5 g sodium chloride/day)	C	1
	Low protein intake: 0.8 g/kg/day in adults with diabetes or without diabetes and CKD (G4–G5), with appropriate education	C (diabetes-related CKD); B (nondiabetic CKD)	2
Smoking	Smoking cessation	Not graded	1
Exercise	Encourage 30–60 min of aerobic exercise at least 5 times/wk	D	1
Proteinuria/albuminuria	Monitoring and follow up; treatment with ACE inhibitors/ARBs, with proteinuria >300 mg/24 h	B	1
Blood pressure	<130/80 mm Hg (diabetes or proteinuric CKD); ^a <140/80 mm Hg (nondiabetic or nonproteinuric CKD)	B	1
Diabetes	HbA _{1c} <7% and use of newer agents (i.e., SGLT2 inhibitors may yield significant benefits for patients with CKD stages 1–4 with regard to CV and kidney outcomes)	A	1
Dyslipidemia	Use of lipid-lowering medications ^b	Not graded	
Metabolic acidosis	Bicarbonate supplementation with levels <20 mEq/L		
Other metabolic risk factors (elevated uric acid)	Insufficient evidence to support or refute the use of agents for hyperuricemia	Not graded	
Multifactorial risk modification/CV risk reduction	Multifactorial intervention strategy addressing BP control and CV risk (with secondary prevention measures, ASA, β -blockers, when appropriate); use of ACE inhibitors or ARBs, statins, and SGLT2 inhibitors where clinically indicated and appropriate	Not graded	

Strength of recommendation: 1, recommended for most patients; 2, suggested for the majority of people, but different choices will be appropriate for some patients; not graded: left to the discretion of care provider, guidance is based on common sense or the given topic does not allow adequate application of evidence.

Certainty of evidence: A, high; B, moderate; C, low; D, very low.

^aUse of ACE inhibitors/ARBs recommended.

^b“Fire-and-forget” strategy: a high-dose statin or moderate dose statin combined with ezetimibe recommended for all CKD patients aged >50 years regardless of serum lipid levels. Patients aged 18–50 years should be treated if they have established CV disease, diabetes, or an estimated 10-year risk of coronary death or nonfatal MI >10%.

ASA, Acetylsalicylic acid; BMI, body mass index; ACE, angiotensin-converting enzyme; ARB, angiotensin-receptor blocker; BP, blood pressure; CKD, chronic kidney disease; CV, cardiovascular; HbA_{1c}, hemoglobin A_{1c}; MI, myocardial infarction; SGLT2, sodium-glucose cotransporter-2.

cotransporter-2 [SGLT2] inhibitors, diuretics, nonsteroidal antiinflammatory drugs [NSAIDs])—are at increased risk of acute-on-chronic kidney disease. Such drugs should be temporarily discontinued during acute intercurrent illnesses, particularly when there is evidence of volume depletion. This advice, termed “sick day rules,” now appears in many CKD management tools (guidelines and pathways).¹²

For decades, BP control and RAAS inhibition have been the cornerstones for preventing progression of CKD. More recently, exciting new evidence has emerged concerning the benefits of SGLT2 inhibitors.^{13,14} Two landmark studies have demonstrated the efficacy of these agents on adverse health outcomes (kidney failure, cardiovascular [CV] events, and mortality) among patients with CKD, irrespective of the presence or absence of diabetes. Thus, slowing progression of CKD in the current era will depend on BP control, RAAS inhibition, and use of SGLT2 inhibitors supplemented by nontherapeutic (lifestyle) and other therapeutic strategies, such as lipid-lowering therapy (Fig. 82.5).

Risk Factor Modification: Lifestyle Interventions

Lifestyle modifications such as dietary changes, exercise, weight loss, and smoking cessation are recommended to prevent CVD among CKD patients. These modifications may prevent progression of CKD, reduce risk of attendant complications, and improve patients' quality of life and survival, although evidence is based on observational studies or small clinical trials.¹⁵

High salt intake and impaired sodium handling by the kidneys are common in CKD and can lead to high BP. Data on the role of salt intake in the progression of kidney disease and risk of CVD in CKD patients is more limited. No large-scale randomized controlled trials (RCTs) on the influence of dietary sodium restriction have specifically enrolled CKD patients. Dietary salt restriction improves some risk factors associated with CKD progression, such as hypertension and albuminuria, and enhances long-term effects on renal function and reduce CV risk.¹⁶ Furthermore, the Dietary Approaches to Stop Hypertension (DASH) study showed the beneficial effects of dietary salt restriction on BP control, which should translate into lower risk of CKD progression.¹⁷ Guidelines recommend lowering sodium intake to less than 90 mmol (<2 g) per day (5 g of salt) in adults (see Chapter 36).

Numerous studies have examined the effects of weight loss (dietary or pharmacologic intervention, or bariatric surgery) on BP.¹⁸ Nevertheless, there is no solid evidence demonstrating a beneficial effect of weight loss per se (i.e., independent of effects on systemic BP and/or diabetes) on progression of CKD.¹⁹

Exercise decreases the risk of incident CV events in numerous patient populations, although not in CKD populations specifically. Exercise can also lower BP, reportedly to a similar extent as a normal dose of an antihypertensive medication. However, adequately powered RCTs reporting the independent effects of exercise on the progression of kidney disease are lacking. One small trial ($n = 30$) randomized CKD

General Guide to Frequency of Monitoring by GFR and Albuminuria Categories

				Persistent albuminuria categories: Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <30 mg/mmol	30–300 mg/g 3–30 mg/mmol	>300 mg/g >30 mg/mmol
GFR categories (mL/min/1.73 m ²): Description and range	G1	Normal or high	≥90	1 if CKD	1	2
	G2	Mildly decreased	60–89	1 if CKD	1	2
	G3a	Mildly to moderately decreased	45–59	1	2	3
	G3b	Moderately to severely decreased	30–44	2	3	3
	G4	Severely decreased	15–29	3	3	4+
	G5	Kidney failure	<15	4+	4+	4+

Fig. 82.4 General guide to frequency of monitoring by glomerular filtration rate (GFR) and albuminuria categories. Numbers represent the frequency of visits for kidney care annually. For example, for CKD stage G5 A1, 4+ represents recommended follow-up visits (as deemed clinically appropriate) at least four times or more per year to receive kidney care. CKD, Chronic kidney disease. (From KDIGO Chronic Kidney Disease Guidelines Working Group. *Kidney Int Suppl.* 2013; 3:1–150.)

BP Target in Comparison With Other Guidelines

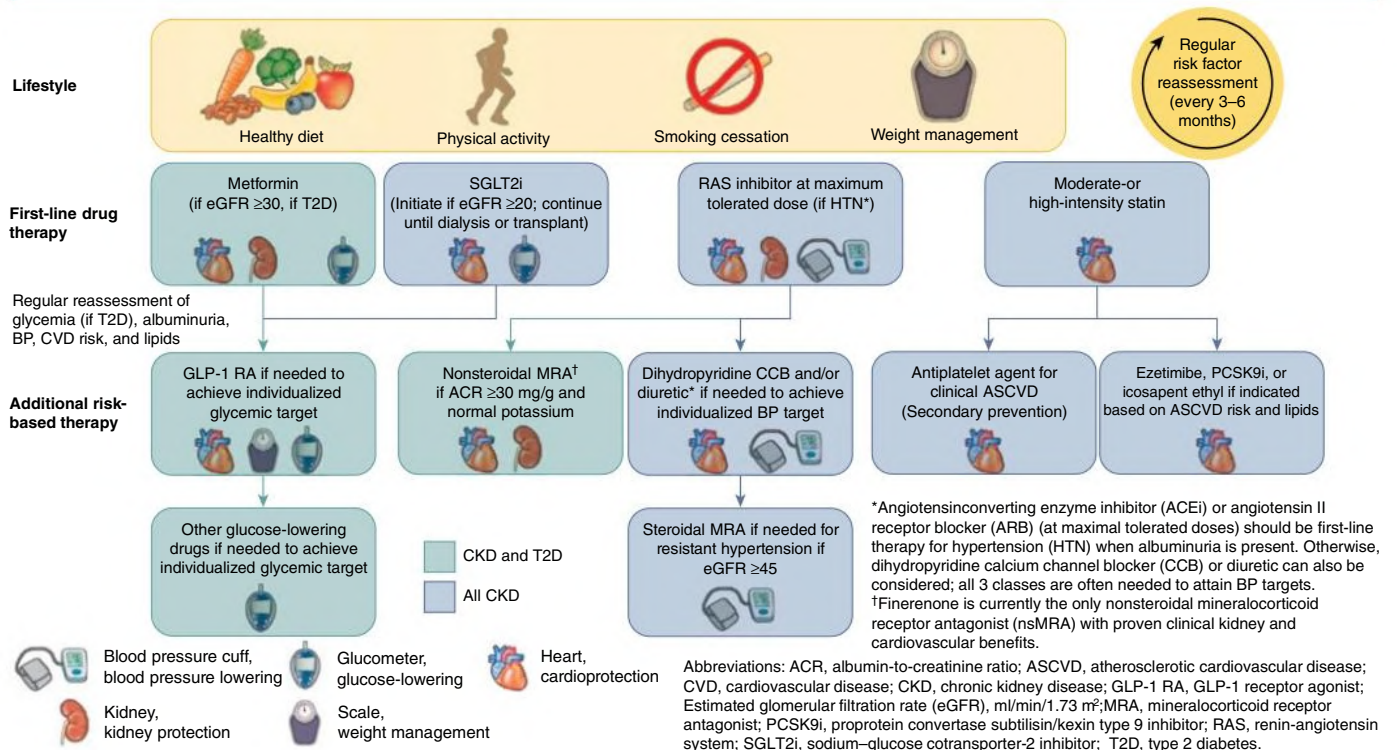


Fig. 82.5 Generic approach to the management of chronic kidney disease, including lifestyle and therapeutic interventions targeted to reduce risk of progression to kidney failure and other adverse health outcomes including cardiovascular events and risk of death. (Modified from de Boer IH, Khunti K, Sadusky T, et al. Diabetes management in chronic kidney disease: a consensus report by the American Diabetes Association [ADA] and Kidney Disease: Improving Global Outcomes [KDIGO]. *Kidney Int.* 2022. Epub ahead of print.)

patients to 30 minutes of daily aerobic exercise or usual care but did not show an effect on the progression of CKD.²⁰ Another observational study showed a 2.8% difference in eGFR decline between physically active people versus those who are sedentary.¹⁸

The association between cigarette smoking and more rapid progression of CKD has been repeatedly shown. A large population-based study in Norway (HUNT II) revealed smoking as a significant risk factor for incident kidney failure and showed that smoking cessation ameliorated this risk, at least among men. Although no RCTs have demonstrated that smoking cessation reduces the risk of progressive CKD,²¹ it is expected that cessation of smoking will improve BP control and lower the risk of adverse kidney and CV outcomes. Thus, smoking cessation should be an integral component of strategies aimed at slowing CKD progression.²¹

In summary, lifestyle interventions are an important adjunct for preventing CKD progression, and guidelines recommend that people with CKD engage in physical activity at levels compatible with their CV health and tolerance (aiming for at least 30 minutes of aerobic exercise 5 times per week) and stop smoking. Moreover, CKD patients should receive expert dietary advice tailored to the severity of their CKD, and modify their diet (particularly salt intake) accordingly.

Risk Factor Modification: Therapeutic Interventions

Blood Pressure Control: Evaluation, Target, and Treatment

BP can be assessed using in-office manual or multiple in-office automated, home, and ambulatory measurements. In practice, BP is typically monitored via in-office manual measurements. Poorly controlled BP is associated with rapid progression of CKD to kidney failure and high risk for CV morbidity and mortality,²²⁻²⁴ but unfortunately a relatively small proportion of patients with CKD meet the recommended BP targets.²⁵ Because CKD is accompanied by an increased risk of CVD, treatment of hypertension in patients with CKD serves two goals: to decrease the risk of CVD and to halt the progression of CKD. Based on new evidence, BP targets for treatment of hypertension in patients with CKD have been lowered in recent years to less than 130/80 mm Hg for those with albuminuria and/or diabetes and less than 140/90 mm Hg for those without diabetes or albuminuria in most guidelines.²⁶ The risk/benefit ratio of intensive BP reduction in patients with CKD continues to be debated. Details on BP targets can be found in [Chapter 37](#).

Ample evidence reveals that lowering BP reduces the risk of major CV events among patients with and without reduced kidney function,²⁷ and this is irrespective of the type of BP-lowering agent, unlike with renoprotection. A meta-analysis of 18 RCTs shows that compared with less intensive BP reduction, intensive BP lowering is associated with lower mortality among patients with CKD stages 3 to 5.²⁵ In a subgroup analysis of 2646 CKD patients in the Systolic Blood Pressure Intervention Trial (SPRINT), outcomes for an intensive treatment group (target systolic BP <120 mm Hg) were compared with outcomes for a standard treatment group (<140 mm Hg) for a median follow-up period of 3.3 years.²⁸ The intensive treatment group had a lower rate of all-cause mortality and a higher rate of AKI as an adverse clinical event. In the main analysis of 9361 patients in SPRINT, risk factors for AKI were older age, non-White ethnicity, lower baseline eGFR, and presence of CVD at baseline.

In terms of renoprotection, overwhelming evidence accumulated over the last 25 years shows that BP control reduces the progression of CKD and supports the use of antihypertensive therapy, specifically with angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) in patients with CKD ([Table 82.3](#) and [Fig. 82.6](#)). In 1993 the first large RCT to yield supporting evidence showed

that ACE inhibitors (captopril) decreased the likelihood of doubling of serum creatinine or kidney failure by 48% in patients with diabetic nephropathy over a 3-year follow-up period, compared with placebo.²⁹ This study was followed by the Irbesartan Diabetic Nephropathy Trial (IDNT), which showed that ARBs (irbesartan) significantly reduced the rate of CKD progression compared with placebo or amlodipine and also affected all-cause mortality.³⁰ Similar findings were reported in the Reduction of Endpoints in NIDDM With the Angiotensin II Antagonist Losartan (RENAAL) trial, in which losartan reduced the risks of a composite kidney outcome (doubling of serum creatinine or kidney failure) and mortality among patients with diabetic nephropathy, compared with placebo.³¹

Studies involving nondiabetic patients also show that lowering BP slows the progression of kidney disease. The first such study was the African American Study of Kidney Disease and Hypertension (AASK) trial that compared intensive and less intensive BP-lowering treatment targets among Black persons with CKD attributed to hypertensive nephrosclerosis.³² Although this study did not find that more intensive BP lowering reduces the risk of death or of progressive CKD, it showed that ACE inhibitors had favorable benefits in slowing eGFR decline compared with β -blockers or calcium channel blockers among the Black population. The Ramipril Efficacy In Nephropathy (REIN) study addressed a similar question in proteinuric, nondiabetic nephropathy³³ and again demonstrated that ACE inhibitors decrease progression to kidney failure compared with other antihypertensive therapies, even in the setting of a similar BP control.

In SPRINT, the composite outcome of 50% or greater decrease in eGFR from baseline or kidney failure did not differ between the intensive and less intensive therapy groups. Although more intensive BP control was associated with similar risk of adverse events and similar kidney outcomes compared with standard treatment, the lower risk of CVD associated with more intensive control might suggest that a lower BP target in CKD patients is preferable. These findings have resulted in debate about the optimal BP targets for CKD patients. For example, the American College of Cardiology/American Heart Association has lowered the target to 130/80 mm Hg in their guidelines for the management of hypertension. In contrast, the KDIGO-recommended BP target for CKD patients endorses this goal for albuminuric and diabetic patients but remains less than 140/90 mm Hg for the group of patients without proteinuria and/or diabetes. The KDIGO 2021 Clinical Practice Guideline for the Management of Blood Pressure in CKD was published proposing an SBP target of less than 120 mm Hg in situations where standardized office readings are feasible in the management of CKD.³⁴ This recommendation stemmed from the CV and survival benefits with this level of intensive BP control from the SPRINT trial. The uptake of this new treatment target will be affected by the casual nature of BP measurements in practice and the challenges associated with it.³⁴

In addition to controlling hypertension, RAAS inhibition is essential for preventing CKD progression.^{31,35} Most guidelines recommend the use of RAAS inhibitors to control hypertension, reduce proteinuria, and slow the progression of CKD. The consensus seems to be that in diabetic and nondiabetic CKD where significant albuminuria is present, RAAS inhibition is the preferred first-line approach, whereas in nonproteinuric disease, the benefit of RAAS inhibition specifically is less well established.³ It is important to recognize and address common situations that may limit the utilization of RAAS inhibition ([Fig. 82.7](#)).

RAAS inhibitors are shown to be effective in slowing CKD progression, even in later stages. In post hoc analyses of the REIN³³ and RENAAL³¹ studies, the reduction of kidney failure events with ACE inhibitors and ARBs was independent of baseline GFR. A similar finding was seen in patients with CKD stage 4 treated with benazepril, in

which carefully selected patients derived a benefit from ACE inhibitor therapy, even at very low eGFR.³⁶ Taken together, these trials suggest that RAAS inhibition effectively reduces the risk of kidney failure for patients even when CKD is severe.³⁷ This position has been reinforced in guideline recommendations for the management of CKD that in the absence of severe hyperkalemia (potassium >6 mmol/L), AKI, or a decrease in eGFR of more than 30%, these medications should be continued, irrespective of baseline eGFR or stage of CKD. Administration of RAAS inhibitors is associated with lower all-cause mortality and a substantial survival benefit for patients with nondialysis CKD (stages G1–5), even with adjustment for confounders and propensity matching across studies.³⁷ Utilization of ACE inhibitors/ARBs remains suboptimal, and discontinuation rates remain relatively high in clinical practice, representing an opportunity to further reduce the number of people who experience progressive CKD. RAAS inhibitors should only be discontinued in situations involving substantial GFR decline (acute-on-chronic kidney disease), and/or severe hyperkalemia that cannot be readily modified or symptomatic hypotension.

Reduction in Albuminuria and Impact on Progressive Kidney Function Loss

Albuminuria is common but not universal in CKD. It is the earliest marker of glomerular diseases, including diabetic glomerulosclerosis, where it generally appears prior to a reduction in kidney function. It is also associated with hypertension/vascular causes in nondiabetic CKD patients, often after a reduction in kidney function.³⁸ Multiple studies show that proteinuria is associated with a faster rate of progression³⁵ and high risk of kidney failure and CVD,^{39,40} irrespective of the underlying cause of CKD. Conversely, reduction of albuminuria via lifestyle and therapeutic measures is associated with better outcomes; the extent of reduction in albuminuria is often proportional to the benefit accrued by such interventions on CKD progression. The superiority of RAAS inhibition compared with other antihypertensive treatments is due at least in part to its efficacy in reducing albuminuria (see Chapter 81).

Recent studies show that SGLT2 inhibitors reduce the risk of adverse outcomes in patients with CKD (see Table 82.3). In clinical trials involving patients with type 2 diabetes mellitus, these agents

TABLE 82.3 Recommended Medications in the Management of Chronic Kidney Disease

Category	Drug	Dose	Adjustment Based on Level of Kidney Function
ACE inhibitors	Benazepril	10–80 mg once daily	eGFR ≥30 mL/min: no dosage adjustment needed eGFR <30 mL/min: reduce initial dose to 5 mg PO once daily
	Captopril	12.5 mg to 50 mg 2–3 times/day (up to 450 mg/day)	eGFR 10–50 mL/min: administer 75% of normal dose every 12–18 hours
	Enalapril	5 mg once daily	eGFR ≤30 mL/min: in adult patients, reduce initial dose to 2.5 mg PO once daily
	Fosinopril	10–80 mg once daily	No dosage adjustment necessary
	Lisinopril	10–40 mg once daily	eGFR 10–30 mL/min: reduce initial recommended dose by 50% for adults (maximum: 40 mg/day)
			eGFR <10 mL/min: reduce initial dosage to 2.5 mg PO once daily
	Perindopril	2–8 mg once daily	Not recommended when eGFR <30 mL/min
	Quinapril	10–80 mg once daily	eGFR 61–89 mL/min: start at 10 mg once daily eGFR 30–60 mL/min: start at 5 mg once daily eGFR 10–29 mL/min: start at 2.5 mg once daily
	Ramipril	2.5–20 mg once daily	Administer 25% of normal dose when eGFR <40 mL/min
	Trandolapril	1–4 mg once daily	eGFR <30 mL/min: reduce initial dose to 0.5 mg/day
ARBs	Azilsartan	20–80 mg once daily	Dose adjustment not required in patients with mild to severe kidney impairment or kidney failure
	Candesartan	8–32 mg once daily	Use with caution in patients with eGFR <30 mL/min
	Irbesartan	75–300 mg once daily	No dosage adjustment necessary
	Losartan	50–100 mg once daily	No dosage adjustment necessary
	Oltmesartan	20–40 mg once daily	No initial dosage adjustment recommended for patients with moderate to significant reduction in kidney function kidney impairment (eGFR <40 mL/min)
	Telmisartan	40–80 mg once daily	No dosage adjustment necessary
	Valsartan	80–320 mg once daily	Use with caution in eGFR <30 mL/min
SGLT2 inhibitors ^a	Canagliflozin	100–300 mg once daily	No dose adjustment if eGFR >60 mL/min/1.73 m ² 100 mg daily if eGFR 30–59 mL/min/1.73 m ² Avoid initiation with eGFR <30 mL/min/1.73 m ² Discontinue when initiating dialysis
	Dapagliflozin	5–10 mg once daily	Initiate at 10 mg/day if eGFR ≥20 mL/min/1.73 m ² Avoid initiation with eGFR <25 mL/min/1.73 m ² Discontinue when initiating dialysis
	Empagliflozin	10–25 mg once daily	Initiate at 10 mg/day for eGFR ≥20 mL/min/1.73 m ² Avoid initiation with eGFR <20 mL/min/1.73 m ^{2a}

^aThese recommendations may change with emerging evidence.

ACE, Angiotensin-converting enzyme; ARB, angiotensin-receptor blocker; eGFR, estimated glomerular filtration rate; PO, orally; SGLT2, sodium-glucose cotransporter-2.

Modified from Navaneethan SD, Zoungas S, Caramori ML, et al. Diabetes management in chronic kidney disease: synopsis of the 2020 KDIGO clinical practice guideline. *Ann Intern Med.* 2020;10.

Initial Antihypertensive Therapy

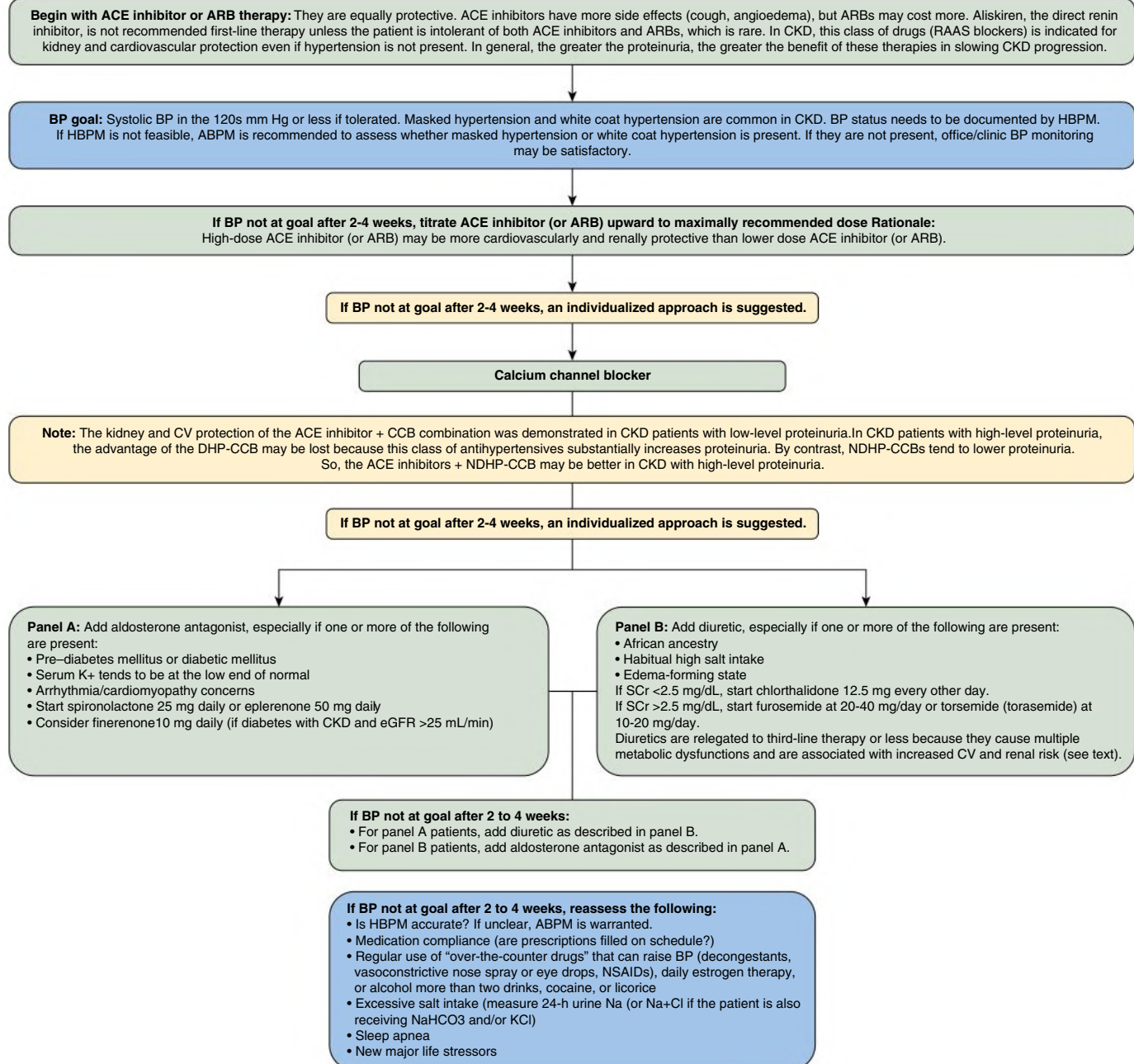


Fig. 82.6 Initial antihypertensive therapy. *ABPM*, Ambulatory blood pressure monitoring; *ACE*, angiotensin-converting enzyme; *ARB*, angiotensin receptor blocker; *BP*, blood pressure; *CKD*, chronic kidney disease; *CV*, cardiovascular; *DHP-CCB*, dihydropyridine calcium channel blocker; *HBPM*, home blood pressure monitoring; *NDHP-CCB*, nondihydropyridine calcium channel blocker; *NSAID*, nonsteroidal antiinflammatory drug; *RAAS*, renin-angiotensin-aldosterone system; *SCr*, serum creatinine; *uACR*, urinary albumin-to-creatinine ratio; *uPCR*, urinary protein-to-creatinine ratio.

reduced albuminuria by 30% to 50% and the incidence of composite hard kidney outcomes by 40% to 50%. Furthermore, they appear to interfere with the major mechanism of proteinuric CKD progression (i.e., glomerular hypertension and hyperfiltration). The Canagliflozin and Renal Events in Diabetes with Established Nephropathy Clinical Evaluation (CREDENCE) study involving 4401 diabetic patients with CKD showed that canagliflozin substantially reduces the risk of progression to kidney failure or mortality associated with kidney or CVD causes.¹³ Recently, the Study to Evaluate the Effect of Dapagliflozin

on Renal Outcomes and Cardiovascular Mortality in Patients With Chronic Kidney Disease (DAPA-CKD) involving 2510 diabetic patients and 1803 nondiabetic patients with CKD showed a significant reduction in the risk of CKD progression, kidney failure, or mortality from kidney or CVD causes.¹⁴ The benefits of these agents appear to be similar in patients with or without diabetes-related CKD, including patients with chronic glomerulonephritides. It is important to consider the baseline level of kidney function before these agents are initiated (Fig. 82.8).^{41,42}

Monitoring of Serum Creatinine and Potassium During ACEi or ARB Treatment: Dose Adjustment and Monitoring of Side Effects

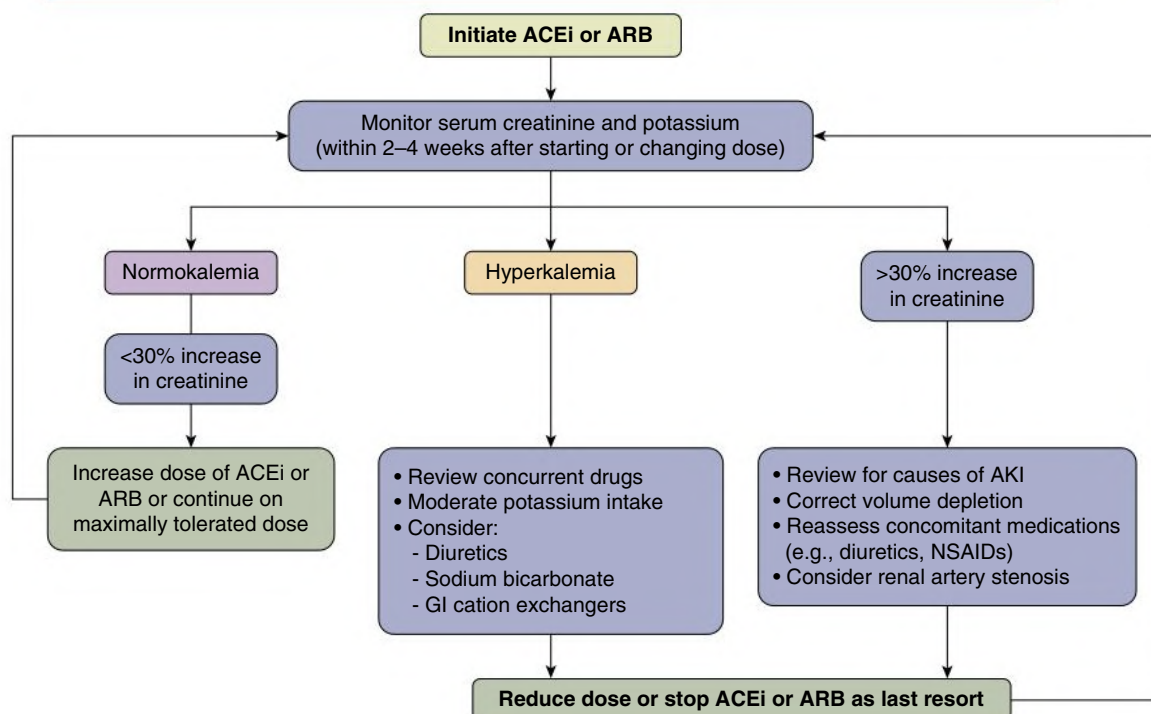


Fig. 82.7 Monitoring of serum creatinine and potassium during angiotensin-converting enzyme inhibitor (ACEi) or angiotensin receptor blocker (ARB) treatment: dose adjustment and monitoring of side effects. AKI, Acute kidney injury; GI, gastrointestinal; NSAID, nonsteroidal antiinflammatory drug. (Modified from Navaneethan SD, Zoungas S, Caramori ML, et al. Diabetes management in chronic kidney disease: synopsis of the 2020 KDIGO clinical practice guideline. *Ann Intern Med.* 2021;174[3]:385–394.)

Initiation of SGLT2i in Patients With CKD (With or Without Diabetes)

	Stage 3b (eGFR 30–44 mL/min/1.73 m ²)	Stage 4 (eGFR 15–29 mL/min/1.73 m ²)	Stage 5 (eGFR <15 mL/min/1.73 m ²)
SGLT2 inhibitors*			
Canagliflozin	Maximum 100 mg daily	Initiation not recommended; may continue 100 mg daily if tolerated for kidney and CV benefit until dialysis	
Dapagliflozin	10 mg daily [†]		Initiation not recommended with eGFR <25 mL/min/1.73 m ² ; may continue if tolerated for kidney and CV benefit until dialysis
Empagliflozin	10 mg daily [†]		Initiation not recommended with eGFR <20 mL/min/1.73 m ² ; may continue if tolerated for kidney and CV benefit until dialysis
Ertugliflozin	Use not recommended with eGFR <45 mL/min/1.73 m ²		

Fig. 82.8 Initiation of sodium-glucose cotransporter-2 inhibitor (SGLT2i) in patients with chronic kidney disease (CKD). *Glucose-lowering efficacy is reduced with SGLT2 inhibitor (SGLT2i) as eGFR declines, but kidney and cardiovascular benefits are preserved. [†]Dapagliflozin is approved for use at 10 mg once daily with an eGFR of 25 to <25 mL/min/1.73 m². [‡]Initiation not recommended with eGFR <30 mL/min/1.73 m² for glycemic control or <20 mL/min/1.73 m² for HF. Higher dose can be used but is not effective for glucose lowering and does not offer further clinical benefit in this range of eGFR. CV, Cardiovascular; DPP-4, dipeptidyl peptidase 4; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HF, heart failure; SGLT2, sodium-glucose cotransporter 2. (Modified from de Boer IH, Khunti K, Sadusky T, et al. Diabetes management in chronic kidney disease: a consensus report by the American Diabetes Association [ADA] and Kidney Disease: Improving Global Outcomes [KDIGO]. *Kidney Int.* 2022. Epub ahead of print.)

In patients treated with RAAS inhibitors (ACE inhibitors or ARBs), aldosterone breakthrough occurs that ameliorates by using aldosterone antagonists (spironolactone, eplerenone) added to ACE inhibitors or ARBs.⁴³ This strategy reduces proteinuria, risk of CKD progression, and CVD, particularly in patients with diabetes.⁴⁴ A recent Cochrane Review demonstrated the benefits of aldosterone antagonists in reducing both albuminuria and BP in patients with mild to moderate CKD; however, the use of these agents was linked to increased risk of hyperkalemia, AKI, and gynecomastia (common with the use of spironolactone [Aldactone]), particularly when combined with RAAS inhibition.⁴⁵

Although earlier guidelines recommended tighter BP control to less than 125/75 mm Hg for patients with albuminuria greater than 1 g/24 h (based on weak evidence from the MDRD trial⁴⁶), this approach is no longer recommended.

Dyslipidemia

Dyslipidemia is often viewed as a consequence of CKD, but experimental data suggest that lipid abnormalities also could accelerate the progression of CKD.⁴⁷ Experimental data show that statins may have synergistic effects with RAAS inhibitors in slowing CKD progression.⁴⁸ Clinical data also show an association between dyslipidemia and a decline in GFR, and that lipid reduction with treatment may preserve kidney function.⁴⁹ Beyond their effects on cholesterol, statins have additional (“pleiotropic”) effects, including antiinflammatory and antifibrotic influences that could theoretically be of benefit in slowing progression of CKD. Small clinical studies suggested a clinical benefit of statin treatment for progressive kidney function loss, and large meta-analyses of patients with stage 3 CKD and coronary disease found small reductions in the rate of kidney function loss over 5 years. However, these potential benefits were not supported in the most rigorous RCT to date, the Study of Heart and Renal Protection (SHARP), which found that combination treatment with simvastatin and ezetimibe did not reduce the risk of progressive kidney function loss or incident albuminuria.^{50,51} However, statins have been consistently shown to reduce mortality and major CV events by 20% in people with CKD not requiring dialysis.⁵¹ Thus, KDIGO guidelines recommend initiation of treatment with a statin or statin/ezetimibe combination in adults 50 years or older with eGFR less than 60 mL/min/1.73 m² but not kidney failure,⁵² irrespective of baseline cholesterol level (a “fire-and-forget” approach). Statin initiation is also recommended for adults with CKD (aged 18–49 years) if they have known coronary artery disease, diabetes mellitus, a history of ischemic stroke, or a greater than 10% estimated 10-year risk of coronary death or non-fatal myocardial infarction. Overall, the indication for statin treatment is prevention of CVD events rather than preventing progressive CKD.

Glycemic Control

Diabetes-related CKD (diabetic nephropathy) is the leading cause of kidney failure worldwide and is considered a microvascular complication of diabetes. Substantial evidence shows that effective management of diabetes ameliorates risk of kidney failure and incident CV events (see Chapter 32). In 2020 KDIGO released clinical practice guidelines for the management of patients with diabetes and CKD, consisting of 12 recommendations and 48 practice points addressing various facets of care, including glycemic monitoring and targets, antihyperglycemic therapies, and educational and integrated care.⁵³

Other Measures

Infection Risk and Prevention

Immunizations against common infections are recommended for patients with CKD, because patients with CKD are at greater risk of

infection and related complications. In addition to routine vaccinations, CKD patients should carefully consider the risks and benefits of vaccinations for influenza, pneumococcal, hepatitis B, zoster, and now COVID-19.⁵⁴ For further details, see Chapters 83 and 87.

Hyperuricemia

Experimental data suggest a link between uric acid concentrations and the progression of hypertensive and toxic nephropathies.⁵⁵ In humans, a link between hyperuricemia and the development of systemic hypertension, CVD, and kidney disease has been postulated.^{55–57} Although recently published studies (PERL and CKD-Fix trials) were important contributions, both showed that urate-lowering therapy does not significantly delay CKD progression in the populations studied, suggesting that lowering uric acid is unlikely to yield benefits for patients with CKD. The major critique of both studies was the fact that neither trial restricted inclusion to people with clinical gout and/or hyperuricemia who are presumably at risk for CKD progression. In fact, individuals with history of gout were excluded, despite the fact that gout affects at least one-third of patients with CKD.⁵⁵

Dietary Interventions

A number of studies investigated the impact of dietary protein restriction on the progression of CKD in the 1970s and 1980s. The MDRD study showed that there was no beneficial effect of dietary protein restriction on progressive kidney function loss.

Avoidance of Nephrotoxins

Chronic lifetime exposure to nephrotoxins such as phenacetin and NSAIDs, heavy metals, and traditional Chinese herbal remedies has been associated with the development of hypertension and CKD progression. For further details, see Chapter 79.

Metabolic Acidosis

A number of studies have linked metabolic acidosis with adverse health consequences in CKD.⁵⁸ The decrease in serum bicarbonate concentration (HCO₃) begins to manifest at lower levels of eGFR (<30 mL/min). This acidosis is linked to progressive kidney function loss and increased mortality risk in CKD.⁵⁸ A few clinical studies have suggested that bicarbonate supplementation may slow the progression of CKD, but the target HCO₃ is unclear, and supplementation is associated with increased pill burden and excess sodium load.^{59,60} We advise bicarbonate treatment when HCO₃ is less than 20 mmol/L, but this is not evidence based.

Multifaceted Strategies

Polypills

Pill burden is high in CKD populations, which may negatively affect quality of life and perhaps reduce the likelihood of adherence. Accordingly, guidelines regarding pharmacologic treatments for patients with CKD recommend combining BP-lowering and lipid-lowering medications with glycemia-lowering medications (for those with diabetes)⁶¹ in a single pill (polypill). These guidelines are similar to those regarding the management of established CVD.

Multidisciplinary Team–Based Care

Multidisciplinary teams (MDTs) have been established in many settings around the world to provide continuing and comprehensive CKD care, particularly for patients with more advanced disease. These teams typically include not only nephrologists but also nurses, surgeons, general practitioners/family physicians, pharmacists, psychotherapists, social workers, nutritionists, and other specialists (e.g., endocrinologists). Emerging evidence on the benefits of MDTs

Management of Common CKD Complications		
Complication	Target	Management
Anemia	Hb 90–110 g/dL	Replace iron deficiency, erythropoietin, or HIF stabilizer
Mineral bone disease	Ca: 2.2–2.35 mmol/L PO ₄ <1.8 mmol/L	Diet/phosphate binders active vitamin D
Metabolic acidosis	Serum bicarbonate >22 mmol/L	Sodium bicarbonate supplementation
Malnutrition	Adequate calorie and protein intake	Dietician support
General health measures: Vaccinations	—	Immunize for influenza and pneumococcus, possibly COVID-19

Fig. 82.9 Management of Common Chronic Kidney Disease (CKD) Complications. Ca, Calcium; Hb, hemoglobin.

BOX 82.1 Referral Criteria

- Acute deterioration of kidney function
- GFR < 30 mL/min/1.73 m²
- Significant and sustained albuminuria (albumin/creatinine ratio ≥ 300 mg/g; equivalent to protein/creatinine ratio ≥ 500 mg/g or proteinuria ≥ 500 mg/24 h)
- CKD progression (sustained decrease in the GFR >5 mL/min/1.73 m² per year or due to a change of category [from G1 to G2, from G2 to G3a, from G3a to G3b, from G3b to G4, or from G4 to G5] when the latter is accompanied by a GFR loss of ≥ 5 mL/min/1.73 m²)^a
- Microhematuria not explained by other causes, sediment with >20 RBCs/field, especially in the case of RBC casts
- Resistant high BP (not controlled with a combination of three antihypertensive drugs, including a diuretic)
- Persistent serum potassium abnormalities
- Recurrent nephrolithiasis
- Hereditary kidney disease

Grade of recommendation: 1, recommendation, evidence B.

^aSmall fluctuations in GFR do not necessarily indicate progression. When the above progression criteria are detected, it is necessary to rule out potentially reversible exacerbation factors (progression vs. exacerbation) such as obstructive uropathy, volume depletion, situations of hemodynamic instability or use of NSAIDs, cyclooxygenase-2 inhibitors, nephrotoxic antibiotics, radiocontrast agents, or renin-angiotensin system blockers in certain hemodynamic conditions.

BP, Blood pressure; CKD, chronic kidney disease; GFR, glomerular filtration rate; NSAIDs, nonsteroidal antiinflammatory drugs; RBCs, red blood cells. From KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int Suppl.* 2013;3.

shows improvements in CKD care, as well as reductions in progressive kidney function loss and hard outcomes such as mortality and CV events.⁶²

MDTs may also enhance the management of other CKD-related complications such as anemia, mineral bone disease, metabolic

acidosis, and electrolyte imbalance that may accelerate loss of kidney function (or interfere with treatments that could prevent CKD, such as RAAS inhibition; Fig. 82.9). Appropriate referrals should be considered when the recommended criteria are met (Box 82.1); see Chapter 83 for details.

SELF-ASSESSMENT QUESTIONS

- Which of the following are modifiable risk factors associated with progression of CKD to kidney failure?
 - Age
 - Race
 - Proteinuria
 - Blood pressure
 - C and D
- Which of the following are adverse health consequences associated with CKD?
 - Heart failure
 - Increased risk of infections
 - Ischemic heart disease and strokes
 - Cognitive dysfunction
 - All of the above
- A 65-year-old patient with a background history of hypertension, ischemic heart disease, and type 2 diabetes mellitus was referred with a serum creatinine of 175 μmol/L (eGFR of 32 mL/min) and a urine albumin-to-creatinine ratio of 300 mg/g (30 mg/mmol). The eGFR was 39 mL/min about a year ago. A diagnosis of CKD (stage G3b:A2) secondary to diabetes and vascular-related causes was made. The current medications include ramipril 5 mg orally daily, hydrochlorothiazide (HCTZ) 25 mg orally daily, atorvastatin 10 mg daily, and linagliptin 5 mg orally daily. The BP is 118/65 mm Hg, and most recent hemoglobin A_{1c} (HbA_{1c}) was 8.3%. You

SELF-ASSESSMENT QUESTIONS—Cont'd

decided to start the patient on canagliflozin at 100 mg orally daily to optimize the glycemia control and to reduce the risk of cardiovascular and kidney disease progression. The patient asked about the specific effects of this new medication on the risk of kidney failure. Which of the following is correct regarding the effects of canagliflozin on kidney function?

- A. Canagliflozin has no effects on kidney function.
 - B. The rate of progression to kidney failure will be amplified over the coming years.
 - C. The risks of kidney stones and urinary tract infections are increased with the use of canagliflozin.
 - D. The risk of proteinuria is increased with the use of canagliflozin.
 - E. There will be an initial decline in kidney function (eGFR), and this will be accompanied by a slower rate of eGFR decline over time and slowing of progression to incident kidney failure.
4. A 60-year-old patient with a long-term history of hypertension, type 2 diabetes, and CKD (stage G3a:A3) is referred for progressive albuminuria. The patient is a current smoker (>20 pack-year history). Over the past year, the urine albumin-to-creatinine ratio has more than doubled from 500 mg/g (50 mg/mmol) to 1200 mg/g (120 mg/mmol). The physical examination was normal except for a slight overweight with body mass index of 30 kg/m². Current medications include irbesartan 300 mg daily, chlorthalidone 25 mg daily, and metformin 500 mg twice daily. BP and diabetes are well controlled and currently at 129/72 mm Hg and HbA_{1c} of 7%, respectively.

In addition to continuing the current medications, which one of the following is the most appropriate management of the increased albuminuria?

- A. Stop smoking
 - B. Recommend exercise and weight loss
 - C. Initiate statin
 - D. Initiate SGLT2 inhibitor
 - E. All of the above
5. You received a call from a general practitioner for advice regarding a 70-year-old patient with a known history of type 2 diabetes (metformin 500 mg twice daily and gliclazide 60 mg daily), hypertension (ramipril 10 mg daily and HCTZ 25 mg daily), and dyslipidemia (atorvastatin 10 mg orally daily). The patient was diagnosed with CKD (Stage G3a:A2) about a year ago. The general practitioner recently started the patient on canagliflozin 100 mg daily. The patient is now suffering from acute gastroenteritis following recent travel. The patient cannot maintain adequate fluid intake and is currently hypotensive from volume depletion upon assessment in clinic. Which of the following medications should be stopped until the patient recovers from the acute illness?
- A. Angiotensin-converting enzyme inhibitors
 - B. Diuretics
 - C. Metformin
 - D. SGLT2 inhibitors
 - E. All of the above

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